

MM Strategy Builder

Django Platform

COMPLETE TECHNICAL ARCHITECTURE REPORT

5 AI MODULES

DJANGO MONOREPO

MARKET MAYA APIS

5

AI Modules

13+

Market Maya APIs

30+

LLM Tools

5

RAG Indexes

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Django 5 · Python 3 · Runware AI (Gemini Flash Lite) · FAISS RAG · Market Maya API

Hosted at http://0.0.0.0:8000 · SQLite DB · SSE Streaming

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1. Core Architecture & Technology Stack

Platform Overview

MM Strategy Builder is a **Django 5 monorepo** running on **http://0.0.0.0:8000**. It hosts five isolated AI-powered strategy builder modules, each with its own URL namespace, templates, RAG index, payload generator, and Market Maya API integration. All modules share a common infrastructure layer for authentication, strategy management, cost logging, and deployment.

Component	Technology / Value
Web Framework	Django 5 (Python 3)
Server	Django dev server — http://0.0.0.0:8000
AI Backend	Runware API (OpenAI-compatible) — google:gemini@3.1-flash-lite
AI Base URL	https://api.runware.ai/v1
Database	SQLite — logs/chat_history.db
Vector Store	FAISS + sentence-transformers (all-MiniLM-L6-v2)
RAG Chunks	k=5 nearest neighbours per query
Streaming	Server-Sent Events (SSE) via Django StreamingHttpResponse
Token Model	Non-native function calling — LLM outputs raw JSON text blocks
Logging	logs/saved_strategies.log (JSON lines) + Django DB (ChatLog model)
Session ID	Hardcoded 'user_1' — no real session management currently

Django Installed Apps & URL Routing

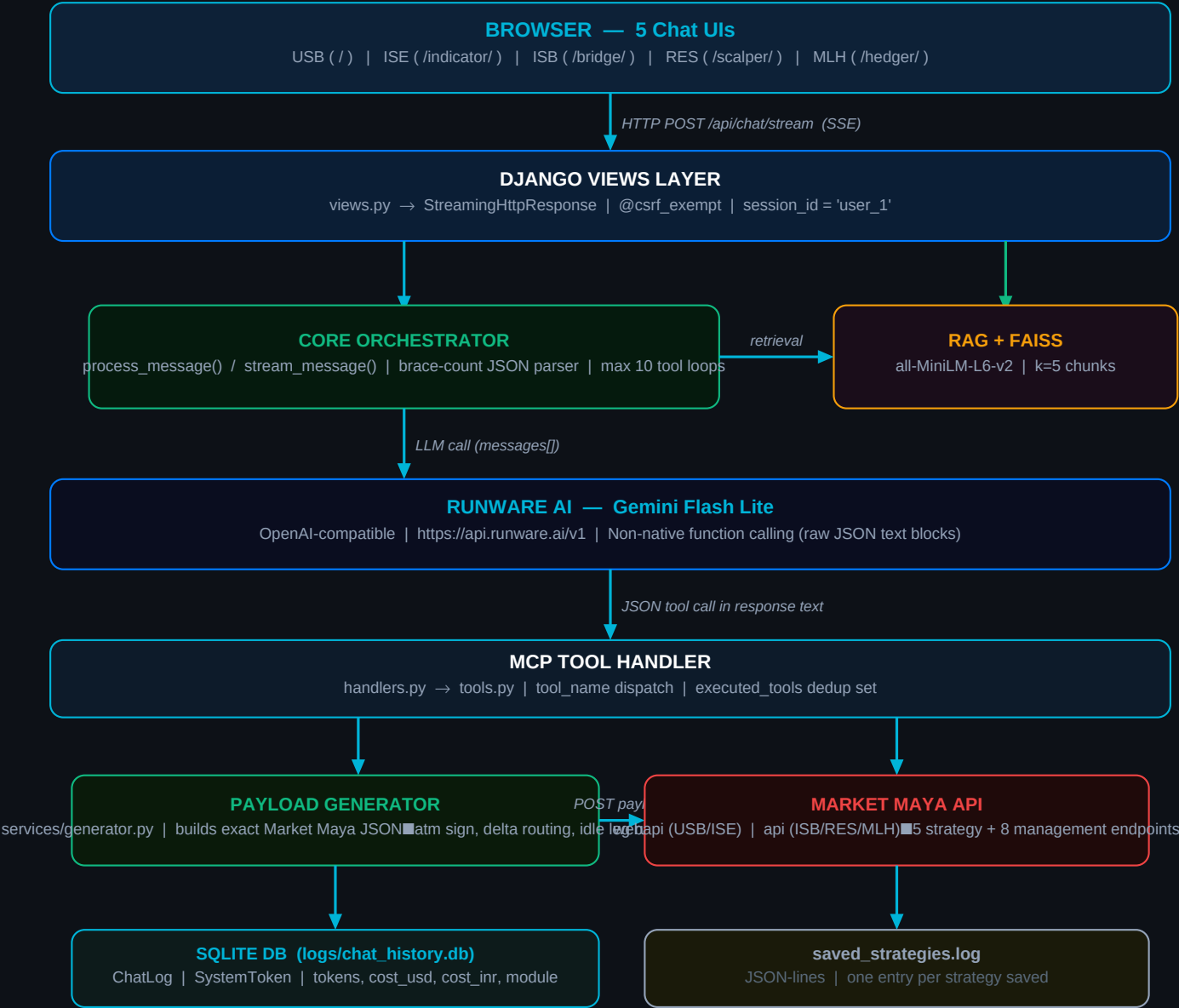
URL Prefix	Module App	Template
/ (root)	Unified_Strategy_Builder	templates/index.html
/indicator/	indicator_engine	templates/indicator_engine.html
/bridge/	inbound_signal_bridge	templates/inbound_signal_bridge.html
/scalper/	rapid_execution_scalper	templates/rapid_execution_scalper.html
/hedger/	multi_leg_hedger	templates/multi_leg_hedger.html
/logs/	chat_logs	templates/chat_logs.html

2. System Architecture Diagram

End-to-End System Architecture

The diagram illustrates the full request path from browser to Market Maya API, including the AI loop, RAG retrieval, payload generation, and persistence layers.

System Architecture — MM Strategy Builder Django | All 5 modules follow identical internal structure



3. Module Overview — The 5 AI Strategy Builders

Module	Route	Market Maya Plugin	API Endpoint (suffix)	Tag
USB — Unified Strategy Builder	/	Multi-leg options (straddles, strangles, spreads, wings, etc.)	api/chat/createtoken/UnifiedStrategy	USB
ISE — Indicator Signal Engine	/indicator/	Indicator-driven (SuperTrend, MACD, RSI, Bollinger Bands)	api/chat/indicator/IndicatorStrategy	ISE
ISB — Inbound Signal Bridge	/bridge/	TradingView / webhook-triggered custom trade strategies	api/chat/custom/CustomTradeStrategy	ISB
RES — Rapid Execution Scalper	/scalper/	Step-based averaging / jobbing (BUY/SELL side, /create/ScalperStrategy)	api/chat/scalper/ScalperStrategy	RES
MLH — Multi-Leg Hedger	/hedger/	Multi-leg w/ Normal / Range Breakout / BTS / STC models	api/chat/multi/MultiLegCallPutStrategy	MLH

ISB, RES, and MLH call `api.marketmaya.com` while USB and ISE call `webapi.marketmaya.com`.

Identical Module Directory Structure

Every module follows the exact same internal layout, ensuring consistency across all five builders:

Path	Responsibility
/	Module root package
views.py	GET (render template) + POST /api/chat + POST /api/chat/stream (SSE)
urls.py	URL patterns for index, chat, stream endpoints
apps.py	Django AppConfig — chat_logs starts token manager daemon here
core/orchestrator.py	AI loop: process_message() + stream_message() with tool parsing
mcp/handlers.py	Routes tool_name → function call
mcp/tools.py	Thin wrappers: calls generator + market_maya + shared services
services/generator.py	Builds exact Market Maya JSON payload from LLM output
services/market_maya.py	POSTs to Market Maya API, logs result to saved_strategies.log
services/validator.py	Validates strategy parameters before generation
rag/retriever.py	FAISS semantic search, k=5, all-MiniLM-L6-v2 embeddings
rag/ingest.py	Builds FAISS index from docs/ markdown files

4. System Data Flow & AI Loop

End-to-End Request Flow

Step	Stage	Detail
1	User Input	Frontend sends POST /api/chat/stream with {message, session_id}
2	View Layer	views.py receives request, calls orchestrator.stream_message()
3	RAG Retrieval	retriever.get_context(user_message) → k=5 FAISS semantic chunks from docs/
4	Message Build	Assemble: [system_prompt, RAG context, chat history, user_message]
5	Confirm Detect	If user said yes/ok/proceed/save → append [SAVE NOW: ...] hint to message
6	LLM Call	POST to Runware API (OpenAI-compatible) with assembled messages
7	Response Parse	Brace-counting extracts complete nested JSON tool call from response text
8	Tool Dispatch	handler.handle_tool_call(tool_name, args) → appropriate service function
9	Generator	services/generator.py builds exact Market Maya JSON payload
10	API Call	services/market_maya.py POSTs payload to Market Maya endpoint
11	Loop / Return	Success → break + return; Error → feed result back, retry (max 10 turns)
12	SSE Stream	Yield {t:chunk} events filtered by brace_depth tracking (hides JSON blocks)
13	Logging	ChatLog DB record saved (tokens, cost_usd, cost_inr); saved_strategies.log updated

Key AI Design Decisions

- **Non-native function calling:** LLM outputs raw JSON text blocks (not OpenAI function-call format). Orchestrator uses brace-counting to extract complete nested JSON objects.
- **Brace-counting parser:** Iterates characters tracking depth: '{' increments, '}' decrements. Extracts first complete JSON object found at depth=0.
- **Infinite-loop prevention:** executed_tools set stores hash(tool_name + sorted_args). Identical tool call detected → skip, preventing same API call twice.
- **SSE streaming filter:** brace_depth variable tracks JSON block boundaries. Characters inside JSON blocks are suppressed from the UI stream.
- **Two-step workflow:** PREVIEW (show 4 Markdown tables, never save) → SAVE (only after explicit user approval). Enforced in system prompt and by [SAVE NOW] hint injection.
- **Confirmation injection:** When confirm words detected (yes/ok/proceed/save), message is augmented with [SAVE NOW: Output ONLY a JSON block calling create_and_save_*] before LLM call.
- **Status events:** {t: 'status', v: '...'} SSE events displayed as inline banners during tool execution.

5. Shared Infrastructure & Services

5.1 Configuration (config.py)

All environment variables and API endpoints are centralised in **config.py** which reads from the **.env** file at startup.

Config Field	Value / Source	Used By
RUNWARE_API_KEY	.env — Runware LLM API key	All orchestrators
RUNWARE_MODEL_ID	google:gemini@3.1-flash-lite	All orchestrators
RUNWARE_BASE_URL	https://api.runware.ai/v1	All orchestrators
MARKET_MAYA_BEARER_TOKEN	.env — JWT bearer token	All Market Maya calls
MARKET_MAYA_EMAIL / PASSWORD	.env — for auto token refresh	token_manager.py
FIREBASE_API_KEY	.env — Firebase auth	token_manager.py
COST_PER_1M_INPUT_TOKENS_USD	0.25 (default)	ChatLog cost calc
COST_PER_1M_OUTPUT_TOKENS_USD	1.50 (default)	ChatLog cost calc
USD_TO_INR_RATE	95.71 (default)	ChatLog cost calc
DOCS_PATH	docs/	RAG ingest
VECTOR_STORE_PATH	<module>/rag/store/faiss_index	RAG retriever

5.2 Market Maya Shared API (services/market_maya_shared.py)

Seven shared functions available in all 5 modules via their MCP tool handlers.

Function	Method	Endpoint	Returns
get_strategies(search, skip, take, trading_type, ids)	POST	api.../getClientMyStrategy	{status, total, strategies[]}
delete_strategy(strategy_id, strategy_name)	POST	api.../delete	{status, message}
get_strategy_record(strategy_id, strategy_name)	GET	webapi.../getCustomTradeRecord	{status, strategy (snake_case)}
modify_strategy(payload)	POST	api.../createCustomTradeStrategy	{status, message}
rename_strategy(strategy_id, strategy_name, new_name)	POST	api.../updateStrategyName	{status, strategy_name, message}
get_balance()	POST	api.../v2/getPointBalance	{status, balance, hold_balance}
_record_to_modify_payload(data)	—	internal	Converts camelCase→snake_case for L

5.3 Deploy Service (services/deploy.py)

Function	Description	Key Parameters
get_deploy_options(strategy_id, strategy_name)	Fetches point balance + live/paper charges. Returns default execution settings	strategy_id: int strategy_name: str
deploy_strategy(id, name, trading_mode, qty_multiply, entry_wait_seconds, exit_wait_seconds)	2-step process: POST /mainStrategy/deploy, POST /mainStrategy/exit	trading_mode: Live/Paper qty_multiply: int entry_wait_seconds: int exit_wait_seconds: int entry/exit_execution_type: PSUEDO entry/exit_wait_seconds: 30

5.4 Database Models (chat_logs/models.py)

Model	Field	Type	Notes
ChatLog	timestamp	DateTimeField(auto_now_add)	When message was sent
ChatLog	module	CharField(10)	USB / ISE / ISB / RES / MLH
ChatLog	session_id	CharField(100, db_index)	Currently hardcoded 'user_1'
ChatLog	user_message	TextField	Full user prompt
ChatLog	ai_response	TextField	Full AI text response
ChatLog	input_tokens	IntegerField(default=0)	LLM prompt tokens
ChatLog	output_tokens	IntegerField(default=0)	LLM completion tokens
ChatLog	total_tokens	IntegerField(default=0)	Sum of input + output
ChatLog	cost_usd	DecimalField(12,8)	Cost in USD
ChatLog	cost_inr	DecimalField(12,4)	Cost in INR
ChatLog	model_used	CharField(100)	Model ID string
SystemToken	key	CharField(50, primary_key)	'market_maya_token'
SystemToken	token	TextField	Current JWT bearer token
SystemToken	firebase_refresh_token	TextField	Long-lived refresh token
SystemToken	expires_at	DateTimeField(null)	JWT expiry from exp claim
SystemToken	updated_at	DateTimeField(auto_now)	Last refresh time

6. USB — Unified Strategy Builder (/)

The most feature-rich module. Builds multi-leg options strategies (straddles, strangles, spreads, iron condors, range breakout plays, BTST/STBT) for Market Maya's Unified Strategy Builder plugin.

6.1 Endpoints

Method	Endpoint	Function	Description
GET	/	index(request)	Renders index.html chat interface
POST	/api/chat	chat(request)	Non-streaming: returns {status, message}
POST	/api/chat/stream	chat_stream(request)	SSE: yields {t,v} events
GET	/api/strategy-counts/	strategy_counts_view(r)	ThreadPoolExecutor fetches all 5 module counts
GET	/api/balance/	balance_view(request)	Proxy to get_balance shared service

6.2 MCP Tools (13 total)

Tool Name	Arguments	Action
get_validation_rules	parameter_name: str	RAG lookup for validation docs
validate_strategy	strategy_json: dict	Runs main + leg validation
generate_payload	strategy_json: dict	Returns Market Maya V3 JSON
deploy	payload: dict	Direct save (raw payload)
create_and_save_strategy	strategy_json: dict	generate_payload → deploy in one step
get_my_strategies	search: str, take: int	List user strategies
delete_strategy	strategy_id or strategy_name	Delete by name or hash ID
get_strategy_record	strategy_id or strategy_name	Fetch full record (snake_case)
modify_strategy	payload: dict	Save changes to existing strategy
rename_strategy	strategy_name, new_name	Rename by name
get_balance	(none)	Fetch account balance
get_deploy_options	strategy_name	Fetch charges before deployment
deploy_strategy	strategy_name, trading_mode, qty_multiply, +L1/L2 fees, +L1/L2 spreads	Full deployment

6.3 Payload Generator — generate_v3_payload()

Converts the LLM's strategy JSON into the exact Market Maya V3 payload. Strategy type ID: **7D0enBHWMRaf4ebeKaB0\$OOMQaC0\$aC0\$**

Field Group	Fields	Notes
Identity	id, strategyName, underlying, mainExchange, mainSegment, mainSymbol	Handle: strip old _NNNN + append time(0%1000)
Trading Mode	isIntraday, productType (MIS/NRML), tradingStartTime, tradingEndTime	Default: intraday=true, MIS
Range Breakout	isRangeBreakOut, rangeEndTime	entry_time doubles as range start time
BTST/STBT	isBtstStbt, btstGapDays	Mutually exclusive with range breakout
Combined Premium	isCombinedPremEntry, totalCombinedPremium	Auto-enables when premium > 0
Master Target	isEnabledMasterTarget, targetBy, intradayTarget, actionOnTarget, noOfIntradayCycle, intradayCycleDelay	
Master Profit Lock	isEnabledProfitLockingTrailing, ifProfitReaches, lockMinimumProfit, increaseInProfitBy, trailProfitBy, noOfTimeTrailTp	noOfTimeTrailTp=9999 (undefaulted) TrailTp
Master SL	isEnabledMasterSI, slBy, intradaySI, actionOnSI, noOfReexecuteOnSI, reexecuteDelayOnSI	
Master SL Trail	isEnabledStoplossTrailing, profitMove, slMove, noOfTrailSI	
Working Days	isEnabledWorkingDays, runMon/Tue/Wed/Thu/Fri/Sat/Sun	isEnabledWorkingDays=true ONLY for non-default
VIX Filter	enableVixFilter, vixStartValue, vixEndValue	
Safety	enableTpSIOnPauseStrategy, sqroffAllLegs, pauseAndSqroffTradingOnMarginExceeded	On Margin Exceeded pause and sqroff

6.4 Per-Leg Payload — `_generate_leg_payload()`

Field Group	Fields	Critical Notes
Identity	id (""), isIdle, tradeSide (BUY/SELL), lot, segment, expiry, openOrderType	openOrderType → idle leg (sequential execution)
Strike Selection	atmType, atm (signed offset), premiumStartRange, premiumEndRange, strikeCondition, strikeDirection (always B or A)	Strike Condition, Strike Direction (always B or A)
Strike Types	Strike By ATM Value / ATM% / Nearest Premium / Premium / Delta / Nearest Delta / Delta Range / Delta Time / Theta P	Delta Range / Delta Time / Theta P
Leg Target	isEnabledLegTarget, targetBy, target, isEnabledActionOnTarget, actionOnTarget, actionOnTargetDelay, actionOnTargetDel	isEnabledActionOnTarget, actionOnTarget, actionOnTargetDelay, actionOnTargetDel
Profit Locking	isProfitLockingAndTrailing, ifProfitReaches, lockMinimumProfit, increaseInProfitBy, trailProfitBy, noOfTimeTrailTp	
Leg SL	isEnabledLegStoploss, slBy, sl, isEnabledActionOnSI, actionOnSI, actionOnSIDelay, noOfReexecuteOnSI, reexecuteDelay	isEnabledActionOnSI, actionOnSI, actionOnSIDelay, noOfReexecuteOnSI, reexecuteDelay
SL Trailing	isEnabledStoplossTrailing, trailSIMarketMove, trailSIMove, noOfTimeTrailSI	
Wait & Trade	isWaitAndTrade, waitFor (Up%/Down%/Up pts/Down pts), waitValue	waitValue=0 when isWaitAndTrade=true
Range Breakout	isExecuteOnRangeBreakout, executeOnRangeBreakout (RangeHigh/Low Break)	RangeHigh/Low Break even if strategy-level RB is on

6.5 Helper Methods

Method	Input → Output	Logic
<code>_parse_margin(value)</code>	str/int → int	Parses '1Cr'→10000000, '1L'→100000, '1K'→1000
<code>_parse_trail_count(value)</code>	str/int → int	'unlimited'/'infinite' or 0 → 9999
<code>_resolve_expiry(expiry, symbol)</code>	str, str → str	NIFTY/SENSEX/BANKEX → Current Week, others → Current Month
<code>_map_condition(value)</code>	str → str	Any / AboveEqual ($\geq$) / BelowEqual ($\leq$)
<code>_map_strike_direction(value)</code>	str → 'BOTH'	Always returns BOTH — direction encoded in atm sign
<code>_map_target_by(value, category)</code>	str, str → str	Money/Point/Point(%) / Range High/Low
<code>_map_wait_direction(value)</code>	str → str	Normalises to: Up % / Down % / Up pts / Down pts

6.6 Critical API Quirks

- *actionOnTargetLegNo stores the DELAY (seconds); actionOnTargetDelay stores the LEG NUMBER — fields are SWAPPED*
- *Reenter Leg requires delay ≥ 5 seconds (actionOnSILegNo ≥ 5) — Market Maya rejects delay=0*
- *isEnabledActionOnTarget requires isEnabledLegTarget=true AND target > 0 — rejected with 400 if no real target*
- *isIdle: true on a leg is sufficient for sequential execution — do NOT combine with action_on_target to trigger it*
- *Nearest Delta / Theta: user value goes into premiumStartRange, NOT the atm field*
- *INDEX segment is always forced to FUT — Market Maya API rejects 'INDEX' as a segment value*
- *isEnabledWorkingDays is true ONLY for non-default day subsets (Mon-Fri default → false)*
- *Range breakout auto-propagates isExecuteOnRangeBreakout=true to all legs EXCEPT idle legs*

7. ISE — Indicator Signal Engine (/indicator/)

Builds indicator-driven strategies. Unique features: backtest via Market Maya WebSocket, indicator master JSON, and **callType** always 'BUY' (direction determined at runtime by signal + isReverseSignal).

7.1 Endpoints

Method	Endpoint	Function
GET	/indicator/	ise_index — renders indicator_engine.html
POST	/indicator/api/chat	ise_chat — non-streaming response
POST	/indicator/api/chat/stream	ise_chat_stream — SSE streaming

7.2 MCP Tools

Tool	Description
create_and_save_ise_strategy(strategy_json)	generate_payload → save in one step
ise_validate_strategy(strategy_json)	Validate chartType, timeFrame, signal, indicators
ise_generate_payload(strategy_json)	Build ISE Market Maya payload
ise_save(payload)	Direct save with raw payload
get_backtest_options(strategy_id/name)	Fetch backtest cost + date range options
run_backtest(strategy_id/name, start_date, end_date)	Initiate WebSocket backtest flow
get_backtest_result(strategy_id/name)	Fetch day/month/year P&L analysis
get_my_strategies / delete / get_record / modify / rename / get_balance	Shared services
get_deploy_options / deploy_strategy	Deploy to Live/Paper trading

7.3 Payload Structure

Field	Value / Notes
strategyTypeId	QFwz7gYjmmabUT8SBvZQGgaC0\$aC0\$
chartType	Candlestick or Heikin-Ashi
timeFrame	5Min / 10Min / 15Min / 30Min / 1Hour / 4Hour / 1Day ONLY
signal	BUY / SELL / Both
entryOrderProduct / exitOrderProduct	MIS (intraday) or NRML (positional)

Field	Value / Notes
underlyingType	Future or Spot/Index
masterTarget / masterSl	Money-based master TP/SL
isTrailSl, profitMove, slMove, noOfTrailSl	Master SL trailing
sqroffBeforeExDays	Days before expiry to square off
weekDays	Array of MON/TUE/WED/THU/FRI/SAT/SUN
callType (per leg)	Always 'BUY' — actual direction from signal + isReverseSignal
isReverseSignal (per leg)	false=follow signal, true=opposite position (hedge legs)
atm (per leg)	CE OTM=+N, PE OTM=-N (signed offset)
indicators[]	Array from indicator_master.json with real Market Maya indicator IDs
LOT_SIZES	BANKNIFTY:30, NIFTY:25, FINNIFTY:40, MIDCPNIFTY:75, SENSEX:20, BANKE:15

7.4 Backtest Service (services/backtest.py)

Note: ISE uses numeric sid for WebSocket messages; hash ID for REST calls.

Step	Action	API / Protocol
1	get_backtest_options: resolve strategy, fetch charge/dates	POST api.../getBacktestOptions
2	Calculate required points: (days) × per_day_charge – free days	Client-side calculation
3	Check point balance vs required_points	GET_BALANCE endpoint
4	Extract client_id from JWT bearer token (base64 decode payload)	Local JWT decode
5	Open WebSocket to algosocketserver.marketmaya.com	WebSocket
6	Send join payload: {token, clientId, userType, executionLevel}	WebSocket message
7	POST deductBacktestPoints: {id, startDate, endDate, executionLevel}	POST api.../deductBacktestPoints
8	Send backteststrategy WS message with strategyId (numeric_sid)	WebSocket message
9	Background thread: keep WS alive 120s, listen for backteststatus	Background daemon thread
10	Return immediately: {status: 'processing', call get_backtest_result after 30s}	REST response
11	get_backtest_result: fetch from 4 REST endpoints (detail/day/month/year)	GET webapi.../

8. ISB — Inbound Signal Bridge (/bridge/)

Builds webhook/TradingView-triggered custom trade strategies. Uses **snake_case** payload (unlike USB/ISE camelCase). Trail SL is locked unless per-leg sl > 0. Exit is in **minutes before market close** (exitMinutes), not absolute time.

8.1 Endpoints

Method	Endpoint	Function
GET	/bridge/	isb_index
POST	/bridge/api/chat	isb_chat
POST	/bridge/api/chat/stream	isb_chat_stream — SSE

8.2 Payload Structure (snake_case)

Field	Value / Notes
strategy_type_id	XBZs7OE0aMivKaB0\$aA0\$Wej3PcwaC0\$aC0\$
product_type	MIS (intraday) / NRML (positional) / CNC
target_by / sl_by	Always 'Money' for master-level
intraday_target / intraday_sl	Master TP/SL in money
exit_minutes	Minutes before market close to exit (NOT absolute time)
capital	Required margin (0 = use live account balance)
allow_update_parameters	Always true
max_position	0 = unlimited concurrent positions
max_position_allocation_percent	1-100%
run_mon/tue/wed/thu/fri/sat/sun	Working days booleans
margin_stock_intraday / stock_positional / futopt_multiplier	Positional multipliers
auto_sqroff_on_contract_exp	Auto square-off on expiry
sqroffAllLegs	Close all legs when any leg exits
sub[] — per leg	exchange, segment, symbol, contract, expiry, option_type, atm, qty, lot, target, sl, trail_sl_market
qty_distribution_options	Fix / Capital(%) / Capital Risk(%) / Allocation Method 1
atm (per leg)	CE OTM=+N, PE OTM=-N (signed offset)

Trail SL per leg requires sl > 0 — locked in UI otherwise. strikePrice: 0 = ATM-relative; non-zero = fixed absolute strike.

9. RES — Rapid Execution Scalper (/scalper/)

Step-based averaging / jobbing strategy. Enters at a base price and averages down/up at fixed intervals. Unique to RES: NIFTY lot size is **65** (vs 25 in ISE/ISB). average_value MUST be > 0.

9.1 Endpoints

Method	Endpoint	Function
GET	/scalper/	res_index
POST	/scalper/api/chat	res_chat
POST	/scalper/api/chat/stream	res_chat_stream — SSE

9.2 Payload Structure

Field	Value / Notes
strategy_type_id	YioJhK5IqBULe8fPLMnXaAaC0\$aC0\$
exchange / segment / symbol / contract / expiry	Main instrument definition
atm / option_type / strike_price	CE OTM=+N, PE OTM=-N; strike_price=0 uses ATM-relative
mix_name	Auto-built: SYMBOL-SEGMENT-CONTRACT-EXPIRY-ATM-OPTION_TYPE
lot / qty	qty = lot × lot_size (NIFTY=65, BANKNIFTY=30, SENSEX=20)
jobbing_side	BUY or SELL
average_by	Point or Percentage — MUST have average_value > 0
average_value	Step size between average entries — NEVER 0
jobbing_start_price / jobbing_end_price	Price range for averaging
maximum_steps	Max number of averaging entries
maximum_target_steps	Steps after which target is triggered
target_by / target / intraday_target	Per-step target (0 = only master TP/SL)
master_tp_money / master_sl_money	Overall strategy target/SL
is_trail_sl	Enable master SL trailing
reset_cycle_by_master_tp_sl	Required when is_trail_sl=true
increase_qty_on_avg / increase_qty / increase_qty_type	Incremental qty per step
scalping_opening_qty	Initial quantity

Field	Value / Notes
is_auto_rollover / rollover_days	Only for positional (not intraday)
intraday_entry_time / intraday_exit_time	Trading window — entry must be before exit
hedge_legs[]	Optional hedge instruments — always qty=0
WEEKLY contracts	Only SENSEX and NIFTY support weekly — others reject

Per-step target = 0 means positions close only on Master TP/SL or end time. Hedge legs always have qty=0.

10. MLH — Multi-Leg Hedger (/hedger/)

Newest module. Supports three trading modes: Normal, Range Breakout, and BTST/STBT. Strategy-level segment is ALWAYS 'FUT' for index underlyings even when all legs are OPT. NIFTY lot size = 65.

10.1 Three Trading Modes

Mode	is_range_break_out	is_btst_stbt	Key Restrictions	Per-Leg Features
Normal	false	false	None — full feature set	Trail SL, Re-entry, Re-execute, Wait & Trade
Range Breakout	true	false	No Re-entry / Re-execute / Wait & Trade. No VTailSL + Sqroff by Fix Time + Condition Ch	TrailSL + Sqroff by Fix Time + Condition Ch
BTST/STBT	false	true	Forces is_intraday=false + NRML. No Trading Cycles	TrailSL + Sqroff by Fix Time + Condition Ch

10.2 Payload Structure

Field	Value / Notes
strategyTypeId	RF8IGNzSfYMaB0\$ENiAa4FpGwaC0\$aC0\$
tradingMode	'Normal' / 'Range Breakout' / 'BTST/STBT'
is_range_break_out / is_btst_stbt	Mutually exclusive booleans (never both true)
underlyingAsset	Built from symbol+segment+exchange
strategy segment	ALWAYS 'FUT' for index underlyings (even when all legs are OPT)
masterTarget / masterSl	0=disabled; masterTargetType: Money/Point/Total Premium(%)
is_trail_sl (master)	Two types: Dynamic or Fix
Dynamic trail	profit_move, sl_move, no_of_trail_sl, start_trail_after_profit, is_live_mtm_profit_move
Fix trail	fix_trail: JSON string '[[profit:N, sl:N], ...]'
start_trail_after_profit	ONLY set when user says 'start trailing AFTER X profit' as a separate threshold
no_of_cycle / cycle_delay	Trading cycles (Normal mode only)
sqroff_by_fix_time + sqroff_time + sqroff_weekly	Additional / BTST-STBT only
sqroff_all_legs	Close all legs when any single leg exits by TP/SL
cosider_closed_pnl	Field has intentional typo (one 'n') — matches Market Maya API exactly
paperTrading / followSimulator	Both hardcoded true (default paper mode)
atm (per leg)	CE OTM=+N, PE OTM=-N. atm_type: 'Fix' (offset) or 'Dynamic' (premium scan)
LOT_SIZES	NIFTY=65, BANKNIFTY=30, FINNIFTY=40, MIDCPNIFTY=75, SENSEX=20, BANKE

10.3 Advance Controls

Feature	Fields	Notes
VIX Filter (Normal only)	enable_vix_filter, vix_start_value, vix_end_value	Only enter when VIX in [start, end]
Dynamic Index (Normal Intraday)	reset_cycle, index_move_by, reset_cycle_index_percent	Adjusts cycle of underlying movement
Sqroff by Fix Time	sqroff_by_fix_time, sqroff_before_expiry_days, sqroff_position, sqroff_week_day	Positional MT Sqroff
Condition Checking (Positional)	chk_con_delay_after_market_start	Minutes to delay TP/SL eval from Day 2
Per-leg ATM Dynamic	atm_type: 'Dynamic', premium_start_range > 0, premium_end_range < 0	Scale premium for premium in range
Per-leg Trail SL	is_trail_sl, trail_sl_by (Point/Money/%), trail_sl_marketmove, trail_sl_slaboverendof_time, trail_sl_cost	Trail SL based on marketmove, trail_sl_slaboverendof_time, trail_sl_cost
Per-leg Re-entry	reentry_on (None/TP Only/SL Only/TP SL Both), no_of_reentry	Normal mode only
Per-leg Re-execute	reexecute_on, no_of_reexecute, reexecute_delay (minutes)	Normal mode only. Re-opens at current ATM after exit.

11. Market Maya API Reference

11.1 Strategy Creation Endpoints

Module	Base URL	Path	Method
USB	webapi.marketmaya.com	/api/mainStrategy/CreateUnifiedStrategy	POST
ISE	webapi.marketmaya.com	/api/mainStrategy/createIndicatorStrategy	POST
ISB	api.marketmaya.com	/api/mainStrategy/createCustomTradeStrategy	POST
RES	api.marketmaya.com	/api/mainStrategy/createScalpingStrategy	POST
MLH	api.marketmaya.com	/api/mainStrategy/CreateMultiLegCallPutStrategy	POST

11.2 Shared Management Endpoints

Purpose	URL	Method
List strategies	api.marketmaya.com/api/V3/mainStrategy/getClientMyStrategy	POST
Delete strategy	api.marketmaya.com/api/mainStrategy/delete	POST
Get full record	webapi.marketmaya.com/api/mainStrategy/getCustomTradeRecord	GET
Modify strategy	api.marketmaya.com/api/mainStrategy/createCustomTradeStrategy	POST
Rename strategy	api.marketmaya.com/api/mainStrategy/updateStrategyName	POST
Get balance	api.marketmaya.com/api/client/v2/getPointBalance	POST
Get charges	api.marketmaya.com/api/subscription/getCharges	POST
Deploy strategy	webapi.marketmaya.com/api/mainStrategy/deploy	POST

11.3 Backtest Endpoints (ISE & MLH)

Purpose	URL	Method
Get backtest options	api.marketmaya.com/api/subscription/getBacktestOptions	POST
Deduct backtest points	api.marketmaya.com/api/subscription/deductBacktestPoints	POST
Get backtest result	webapi.marketmaya.com/api/mainStrategy/getBackTestResult	GET
Get strategy detail	webapi.marketmaya.com/api/mainStrategy/getClientMyStrategyDetail	GET
Get day history	webapi.marketmaya.com/api/mainStrategy/getDayTradeHistory	GET
Get month history	webapi.marketmaya.com/api/mainStrategy/getMonthTradeHistory	GET

Purpose	URL	Method
Get year history	webapi.marketmaya.com/api/mainStrategy/getYearTradeHistory	GET
Backtest WebSocket	wss://algotsocketserver.marketmaya.com/algo-server?usertype=Client&executionLevel=1001%208&id={client_	WS

11.4 Strategy Type IDs

Module	Strategy Type ID
USB — Unified Strategy Builder	7D0enBHWMRaf4ebeKaB0\$OOMQaC0\$aC0\$
ISE — Indicator Signal Engine	QFwz7gYjmmabUT8SBvZQGgaC0\$aC0\$
ISB — Inbound Signal Bridge	XBZs7OE0aMivKaB0\$aA0\$Wej3PcwaC0\$aC0\$
RES — Rapid Execution Scalper	YioJhK5lqBULe8fPLMnXaAaC0\$aC0\$
MLH — Multi-Leg Hedger	RF8IGNzSfYMaB0\$ENiAa4FpGwaC0\$aC0\$

12. Generator Key Quirks & Critical Rules

Mod	Quirk / Rule	Impact
USB	actionOnTargetLegNo stores DELAY, actionOnTargetDelay stores LEG NUMBER swapped	Wrong leg triggered
USB	Reenter Leg / Execute Leg requires delay ≥ 5 (actionOnSILegNo ≥ 5)	delay=0 rejected by Market Maya with 400
USB	isEnabledActionOnTarget requires isEnabledLegTarget=true AND target > 0	'Invalid Action on Target' 400 error
USB	isIdle=true alone is sufficient for sequential leg — no action_on_target needed	Adding action_on_target without target causes 400
USB	Nearest Delta / Theta: user value $\rightarrow$ premiumStartRange (NOT atm field)	Market Maya reads delta/theta from premiumStartRange
USB	PE OTM = negative atm offset (below ATM); CE OTM = positive (above ATM)	Wrong sign $\rightarrow$ wrong strike selected
USB	_map_strike_direction always returns BOTH — direction encoded by signal	OTM/ATM direction field is irrelevant in USB
USB	INDEX segment forced to FUT — Market Maya rejects INDEX	All index derivatives must use FUT segment
USB	isEnabledWorkingDays = true ONLY if user selected non-default days	Generator detects and ignores default Mon-Fri outputs
USB	Range breakout: isExecuteOnRangeBreakout=true auto-set EXCEPT for hedge legs	Hedge legs must have isExecuteOnRangeBreakout=false
USB	Strategy name: strip old _NNNN suffix, append new time(0%10000	Prevents duplicate names on resave
USB	_parse_trail_count(0) $\rightarrow$ 9999 (unlimited)	0 and 'unlimited'/'infinite' all map to 9999
ISE	callType is always BUY — actual direction from signal + isReverseSignal	Never set SELL callType
ISE	Indicator params loaded from indicator_master.json (real Market Maya indicator codes)	Indicate (Do) invent indicator codes
ISE	Supported timeframes ONLY: 5Min/10Min/15Min/30Min/1Hour/4Hour/1Day	Anything else rejected
ISE	WebSocket uses numeric sid for backtest (not hash ID)	Hash ID for REST; numeric sid for WebSocket
ISB	snake_case payload (vs camelCase in USB/ISE)	Different field naming convention
ISB	exitMinutes = minutes before market close (NOT absolute exit time)	Confusable with USB exit_time field
ISB	capital = required margin (0 = use live account balance)	
RES	NIFTY lot size = 65 (not 25 as in ISE/ISB)	Different from other modules
RES	average_value MUST be > 0 — zero is invalid	API rejects average_value=0
RES	WEEKLY contracts only for SENSEX and NIFTY	Use MONTHLY for other instruments
RES	Hedge legs always have qty=0	Non-zero qty rejected for hedge legs
MLH	NIFTY lot size = 65	Same as RES
MLH	cosider_closed_pnl (one 'n') — intentional typo matching Market Maya API	Using 'consider' breaks the API
MLH	Strategy-level segment is ALWAYS FUT for index underlyings	Never change to OPT at strategy level
MLH	paperTrading: true, followSimulator: true hardcoded (default paper mode)	Paper mode is the default

Mod	Quirk / Rule	Impact
MLH	start_trail_after_profit: only set when user says 'start trailing AFTER X profit'	Don't not set just because profit_move is mentioned
All	Strategy name suffix: strip existing _NNNN, append time()%10000	Fresh unique name every save
All	Infinite-loop prevention: executed_tools set with hash(tool_name + sorted_params)	Prevents same API call twice in one session

13. Frontend & SSE Streaming

13.1 SSE Event Protocol

Event Type	Payload	Frontend Action
{t: 'chunk', v: text}	Partial AI response	Append to message div, re-render Markdown
{t: 'status', v: message}	Status string	Show inline status banner
{t: 'done', in_tok: N, out_tok: N}	Token counts	Finalize message, remove cursor, update token display
{t: 'error', v: message}	Error string	Show error in chat

13.2 Key JavaScript Functions (static/js/chat.js)

Function	Description
handleSend()	Fetch /api/chat/stream, parse SSE data: lines, dispatch event types
appendMessage(r, txt)	Append to chat, render Markdown via marked.parse(), auto-scroll
setInputLocked(bool)	Disable/enable send button and input during AI response
loadStrategyCounts()	GET /api/strategy-counts/, update sidebar badges
loadBalance()	GET /api/balance/, display point balance
toggleTheme()	Switch dark/light, persist in localStorage

13.3 CSS Theme System (static/css/style.css)

Variable	Dark Value	Purpose
--bg-dark	#0a0e14	Page background
--card-bg	rgba(30,34,42,0.7)	Chat card background
--primary-glow	#00f2ff	Accent / highlight colour
--accent-blue	#007bff	Buttons and links
--text-main	#e0e6ed	Primary text
--text-muted	#94a3b8	Secondary text
--border-color	rgba(255,255,255,0.1)	Borders
--success	#10b981	Success states
--danger	#ef4444	Error states

14. Chat Logs & Analytics Dashboard (/logs/)

Dashboard at /logs/ provides full analytics across all 5 modules with time filtering, per-module breakdowns, and daily cost charts.

Feature	Details
Filter options	module (USB/ISE/ISB/RES/MLH), quick (today/yesterday/last7/thismonth), date_from, date_to
Stats shown	Total requests, total tokens (in+out), total cost USD & INR
Per-module breakdown	Count, tokens, cost for each of the 5 modules
Daily cost chart	Last 30 days, one bar per day, cost_usd aggregated
Max logs shown	200 in dashboard, 500 in /logs/api/ JSON endpoint
logs_api endpoint	Returns JSON: logs[], totals{} — aggregated via Count + Sum on queryset

15. Authentication & Security

Aspect	Implementation
Market Maya Auth	Static Bearer JWT from .env — all API calls include Authorization header
Auto Token Refresh (planned)	token_manager.py: startup() fetches token at boot, thread refreshes every 30 min
Token Storage (planned)	SystemToken DB model — key='market_maya_token', JWT + firebase_refresh_token + expires_at
Firebase Flow (planned)	signInWithPassword → idToken + refreshToken → googleLogin → Market Maya JWT
CSRF	@csrf_exempt on all POST chat endpoints (SSE requires exemption)
Request Timeouts	30s standard API calls, 180s chat endpoints
CORS	Not explicitly configured — relies on same-origin browser requests
Session	Hardcoded session_id='user_1' — no real user authentication currently

16. Testing Framework

Test File	Endpoint	Tag	What It Tests
tests/run_usb_tests.py	http://localhost:8000/api/chat	USB	Full USB parameter coverage (20 prompts)
tests/run_ise_tests.py	http://localhost:8000/indicator/api/chat	ISE	Indicators, timeframes, signals, backtest
tests/run_isb_tests.py	http://localhost:8000/bridge/api/chat	ISB	Custom trade strategies, trail SL, exit minutes
tests/run_res_tests.py	http://localhost:8000/scalper/api/chat	RES	Scalping params, average, jobbing side
tests/run_mlh_tests.py	http://localhost:8000/hedger/api/chat	MLH	Multi-leg modes, trail types, VIX filter

All tests require the server running (python manage.py runserver 0.0.0.0:8000). Each runner: sends chat POST → reads payload from logs/saved_strategies.log → validates key fields → reports pass/fail. Reports saved in tests/reports/.

17. Cost Calculation & Tokenomics

Cost is calculated per interaction and stored in the ChatLog model:

```
cost_usd = (input_tokens × 0.25 + output_tokens × 1.50) / 1,000,000
cost_inr = cost_usd × 95.71
```

Config Key	Default	Notes
COST_PER_1M_INPUT_TOKENS_USD	\$0.25	Gemini Flash Lite input pricing
COST_PER_1M_OUTPUT_TOKENS_USD	\$1.50	Gemini Flash Lite output pricing
USD_TO_INR_RATE	■95.71	Conversion rate (configurable in .env)
Stored precision	8dp / 4dp	cost_usd: 8 decimal places, cost_inr: 4 decimal places
Dashboard aggregation	Daily totals	Last 30 days, per-module breakdown, all-time stats

18. Running the Project

Action	Command
Start server	<code>python manage.py runserver 0.0.0.0:8000</code>
Run migrations	<code>python manage.py migrate</code>
Rebuild RAG (USB)	<code>python -m Unified_Strategy_Builder.rag.ingest</code>
Rebuild RAG (ISE)	<code>python -m indicator_engine.rag.ingest</code>
Rebuild RAG (ISB)	<code>python -m inbound_signal_bridge.rag.ingest</code>
Rebuild RAG (RES)	<code>python -m rapid_execution_scalper.rag.ingest</code>
Rebuild RAG (MLH)	<code>python -m multi_leg_hedger.rag.ingest</code>
Run USB tests	<code>python tests/run_usb_tests.py</code>
Run ISE tests	<code>python tests/run_ise_tests.py</code>
Run ISB tests	<code>python tests/run_isb_tests.py</code>
Run RES tests	<code>python tests/run_res_tests.py</code>
Run MLH tests	<code>python tests/run_mlh_tests.py</code>

Module Access URLs

Module	URL
USB — Unified Strategy Builder	http://0.0.0.0:8000/
ISE — Indicator Signal Engine	http://0.0.0.0:8000/indicator/
ISB — Inbound Signal Bridge	http://0.0.0.0:8000/bridge/
RES — Rapid Execution Scalper	http://0.0.0.0:8000/scalper/
MLH — Multi-Leg Hedger	http://0.0.0.0:8000/hedger/
Chat Logs & Analytics	http://0.0.0.0:8000/logs/

Required .env Variables

Variable	Required	Description
RUNWARE_API_KEY	✓ Yes	Runware LLM API key
RUNWARE_MODEL_ID	✓ Yes	e.g. google:gemini@3.1-flash-lite
MARKET_MAYA_BEARER_TOKEN	✓ Yes	JWT bearer token (manual or auto-refreshed)
MARKET_MAYA_EMAIL	Optional	For auto token refresh flow
MARKET_MAYA_PASSWORD	Optional	For auto token refresh flow
FIREBASE_API_KEY	Optional	For Firebase-based token refresh